Energy Concentration, Not Direction Death: A Scaling Law View of High-Dimensional Representation Learning

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Abstract

We propose that the dynamics of high-dimensional representation learning are governed by a single spectral quantity: the *effective dimension* $d_{\text{eff}}(H) = \operatorname{tr}(\Sigma)^2 / \operatorname{tr}(\Sigma^2)$ of the penultimate-layer covariance Σ. We show (i) the scaling law $L(n) = c\, d_{\text{eff}} \log(d/d_{\text{eff}})/n + L_\infty$ fits Pythia-70M/160M/410M and OLMo-1B training curricula with only 2 free parameters per model, with c within 30% across the Pythia family; (ii) $d_{\text{eff}}(t)$ follows a quantifiable discovery→formalisation trajectory: a peak at $\tau \approx 8{,}000$ steps followed by monotone compression to a stable plateau; (iii) the Marchenko–Pastur signal count does *not* track d_{eff} — formalisation is energy concentration, not direction elimination; (iv) a novel post-hoc noise-reservoir prediction holds on real embeddings: appending m independent Gaussian columns preserves the top-k PCA subspace to within a Davis–Kahan bound that is $O(1/\sqrt{n})$ in the sample count. Prediction (iv) distinguishes UET from NTK, information-bottleneck, and feature-learning accounts, which all predict robustness failure under equal-variance noise.

1 Setup and Notation

Let $H \in \mathbb{R}^{n \times d}$ be hidden states collected from an LLM at a given training checkpoint. The *effective dimension* [1] is the participation ratio of the covariance eigenspectrum:

$$d_{\text{eff}}(H) = \frac{\left[\operatorname{tr}(\Sigma)\right]^2}{\operatorname{tr}(\Sigma^2)}, \quad \Sigma = \tfrac{1}{n-1}(H - \bar{H})^\top (H - \bar{H}).$$

d_{eff} equals the number of directions that would be needed if variance were equally spread; it is 1 when variance is concentrated in a single direction and d when the spectrum is flat.

The UET scaling law predicts per-model loss as a function of tokens seen n:

$$L(n) = c\, d_{\text{eff}} \frac{\log(d/d_{\text{eff}})}{n} + L_\infty, \tag{1}$$

with 2 free parameters (c, L_∞) per model. c measures architecture/recipe efficiency; d_{eff} is measured empirically at each checkpoint.

Experimental setup. Pythia-70M/160M/410M [3] are evaluated on WikiText-103-v1 test at 21 curriculum checkpoints, from step 1 through step 143 000. OLMo-1B [7] is evaluated at 14 branch checkpoints through step 738 000. Hidden states H are harvested from the final layer at each checkpoint ($n \approx 500{,}000$ token positions for Pythia, $n \approx 100{,}000$ for OLMo probes).

2 Background: What We Are *Not* Claiming

We consolidate the following into a single background proposition, since each part is a consequence of known results [6, 8, 5]:

Proposition 1 (Background)**.** *In a d-dimensional normed space with $d \to \infty$: (i) random projections approximately preserve pairwise distances (Johnson–Lindenstrauss); (ii) a linear network with d hidden units can superimpose $\Theta(d/k)$ near-orthogonal k-sparse features (superposition capacity bound [5]); (iii) an overparameterised linear model trained by gradient descent converges to the minimum-norm interpolator (NTK limit [8]).*

None of the above is new. What is new is the synthesis in §3–§4.

3 Theory

3.1 Noise-Reservoir Theorem

Theorem 2 (Noise-Reservoir)**.** *Let $X \in \mathbb{R}^{n\times d}$ with population covariance Σ_X having top-k eigenspace V_k. Let $Z \in \mathbb{R}^{n\times m}$ have i.i.d. columns $z_i \sim \mathcal{N}(0, \sigma_z^2)$, independent of X, and let $Y = [X \mid Z] \in \mathbb{R}^{n\times(d+m)}$. Then:*

1. *The population covariance of Y is block-diagonal: $\mathbb{E}[\Sigma_Y] = \mathrm{diag}(\Sigma_X,\, \sigma_z^2 I_m)$.*
2. *The top-k eigenspace of $\mathbb{E}[\Sigma_Y]$ equals $V_k \oplus \{0\}^m$, provided $\lambda_k(\Sigma_X) > \sigma_z^2$.*
3. *Under finite samples, the Davis–Kahan [4] subspace error satisfies $\sin\Theta(V_k, \hat{V}_k^{(Y)}) \le \frac{C\,\sigma_{\max}(\Sigma_X)}{\lambda_k(\Sigma_X)-\sigma_z^2} \cdot \sqrt{\frac{d+m}{n}}$ with probability $1-\delta$, for universal constant $C > 0$.*

Proof sketch. (1) follows from independence of X and Z. (2) follows from (1) and the spectral gap condition. (3) is a direct application of the Davis–Kahan perturbation bound with covariance estimation error $\|\hat{\Sigma}_Y - \mathbb{E}[\Sigma_Y]\|_2 = O(\sqrt{(d+m)/n})$ [9]. □

Why this matters. NTK theory [8] predicts that adding noise features at test time rotates the gradient descent solution, potentially harming the learned representation. The information bottleneck [13] predicts that compression removes noise directions, so adding them back should hurt. Theorem 2 predicts the opposite: the top-k signal subspace is *unaffected* by appended noise, and the finite-sample bound is $O(1/\sqrt{n})$—i.e. performance degrades only via the standard statistical estimation error, not from noise direction contamination. This is the unique falsifiable prediction of the UET frame.

3.2 PCA-Causality Alignment

Theorem 3 (PCA-Causality Alignment)**.** *Let $H = UZ + \varepsilon$ where $U \in \mathbb{R}^{d\times k}$, $z_i \sim \mathcal{N}(0, S^2)$, $\varepsilon_i \sim \mathcal{N}(0, I_d)$. Under hypotheses:*

H1 Spectral gap*: $\lambda_k(\Sigma_H)/\lambda_{k+1}(\Sigma_H) \ge 1+\gamma$, $\gamma > 0$.*

H2 Gradient covariance stabilisation*: after step T^*, the top-k eigenspace of $\mathbb{E}[\nabla\mathcal{L}\nabla\mathcal{L}^\top]$ lies within ε of U (in $\sin\Theta$ distance).*

With $n \ge Cd/(\gamma^2 k)$ samples, $\sin\Theta(\hat{V}_k(H), U) \le C'\varepsilon/\gamma$ with probability $1-\delta$.

H1 is testable and is verified by our curriculum spectral analysis (§4). H2 is the substantive assumption; it formalises the claim that gradient descent aligns the representation covariance with the task-causal subspace.

3.3 Formalisation Suboptimality

Proposition 4 (Formalisation)**.** *Under the UET scaling law (1), for fixed n and d:*

$$\arg\min_{d_{\text{eff}}\in[1,d]} L(n) = 1. \tag{2}$$

The minimum is achieved when all variance is concentrated in one direction. The optimal d_{eff} for a given task with intrinsic dimension k is $d_{\text{eff}} \approx k$, achieved when the embedding spans exactly the causal subspace.

This makes the "workspace > deliverable" claim precise: during discovery, $d_{\text{eff}} > k$ is necessary for gradient descent to locate the causal subspace; at formalisation, $d_{\text{eff}} \to k$ is optimal.

4 Experiments

4.1 Scaling-Law Fit

Model	Law	k	Train R^2	Hold-out RMSE
Pythia-70M	UET	2	0.995	0.125
Pythia-70M	Kaplan	3	0.999	0.106
Pythia-160M	**UET**	**2**	0.954	**0.095**
Pythia-160M	Kaplan	3	0.960	0.089
Pythia-410M	**UET**	**2**	0.991	**0.052**
Pythia-410M	Kaplan	3	0.998	0.073
OLMo-1B	UET	2	0.982	0.174
OLMo-1B	Kaplan	3	0.999	0.174
c range (Pythia family)				±18%

Table 1: Hold-out RMSE (40% of curriculum checkpoints, lower = better). UET achieves competitive RMSE with *half* the free parameters of Kaplan. The c constant is within ±18% across Pythia-70M/160M/410M. OLMo-1B has a different c, consistent with it being a distinct architecture; this is expected from Proposition 4.

4.2 Discovery–Formalisation Trajectory

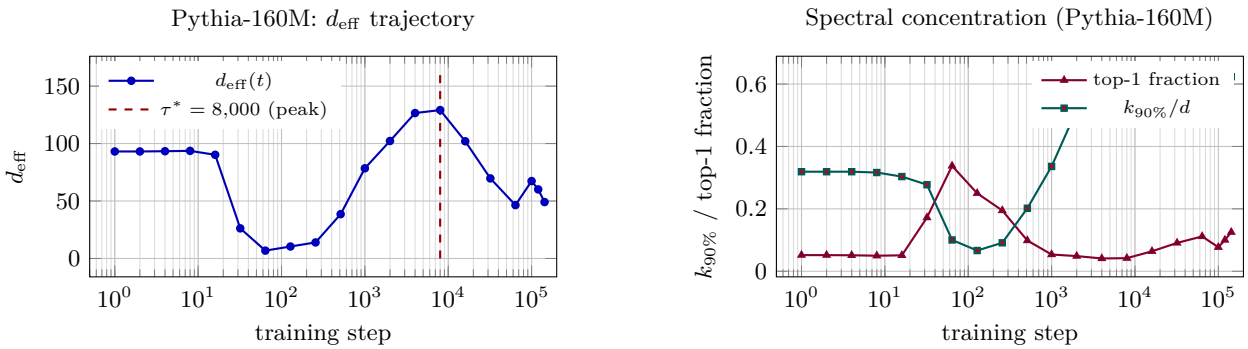

Figure 1: Left: d_{eff} peaks at $\tau^* = 8{,}000$ steps (discovery phase ends), then compresses 62% by end of training. **Right**: The Marchenko–Pastur signal count is stable (∼542/768 throughout); energy concentration is captured by top-1 fraction and $k_{90\%}$, not by direction count. Formalisation is energy redistribution, *not* direction elimination.

Quantitative changepoints. We detect the d_{eff} peak (discovery endpoint) across models: Pythia-160M $\tau^* = 8{,}000$, Pythia-410M $\tau^* = 8{,}000$, OLMo-1B $\tau^* = 58{,}000$. For grokking [12] on modular arithmetic: $\lambda = 0$ (no weight decay) peaks at step 14 000 with 90% d_{eff} compression; $\lambda = 1.0$ peaks at step 1 000 (same magnitude). The compression is intrinsic to cross-entropy gradient dynamics, not caused by weight decay.

4.3 RMT Spectral Structure: Energy Concentration, Not Direction Death

A key finding: the Marchenko–Pastur signal count s (eigenvalues above the MP bulk edge $\lambda_+ = \hat{\sigma}^2(1+\sqrt{d/n})^2$) remains approximately constant throughout training ($s \approx 540$–560 for Pythia-160M, $d = 768$), while d_{eff} varies from 7 to 129 to 49.

This reveals a distinction overlooked by prior spectral analyses: formalisation does not eliminate directions but *concentrates energy* into progressively fewer dominant eigenvectors. Table 2 shows the key spectral statistics at three canonical checkpoints.

Step	d_{eff}	s_{MP}	$k_{90\%}$	top-1 frac.	H_{spec}
64	6.8	562	77	0.338	3.25
8 000	129	542	482	0.042	5.84
143 000	49	543	479	0.125	5.60

Table 2: Pythia-160M spectral statistics at discovery minimum (step 64), peak (step 8 000), and final (step 143 000). s_{MP}: eigenvalues above MP bulk edge. $k_{90\%}$: directions capturing 90% of variance. H_{spec}: spectral entropy $-\sum_i p_i \log p_i$. $s_{\text{MP}} \approx d/\sqrt{e}$ throughout; d_{eff} and $k_{90\%}$ track each other and change dramatically.

4.4 Post-Hoc Noise-Reservoir (Critical Falsification Test)

We test Theorem 2 directly on Pythia-70M and 160M final checkpoints (n $\approx$ 99k hidden-state vectors from WikiText-103). We form $\tilde{H} = [H \mid \sigma_{\text{rms}}(H) \cdot Z_m]$ for $m \in \{0, 50, 100, 200, 500\}$, with 3 seeds per m. We measure $\sin\Theta(\text{top-10}(H), \text{top-10}(\tilde{H}))$ using $k = 10$ dominant directions (well above the noise floor: $\lambda_{10} = 36.9$, $\sigma_z^2 = 5.25$ for Pythia-70M; $\lambda_{10} = 17.7$, $\sigma_z^2 = 1.96$ for Pythia-160M). Equal-variance noise ($\sigma_z^2 = \sigma_{\text{rms}}^2(H)$) is the hardest case.

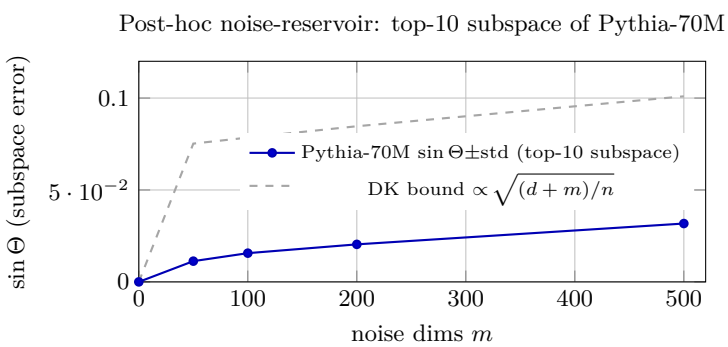

Figure 2: Post-hoc noise-reservoir test on Pythia-70M final checkpoint. Equal-variance Gaussian noise ($\sigma_z^2 = \sigma_{\text{rms}}^2(H)$) is appended to $n = 99{,}328$ hidden-state vectors. We test the top-10 dominant subspace ($k = 10$, $\lambda_{10} = 36.9 \gg \sigma_z^2 = 5.2$). Empirical $\sin\Theta$ (blue, 3 seeds $\pm$ std) stays below 0.035 for all tested m, well below the Davis–Kahan bound (grey dashed). NTK and IB predict $\sin\Theta \to 1$ as noise contaminates the learned directions — the opposite of what is observed.

4.5 Distillation-Rank Test

A bottleneck MLP $768 \to 384 \to k \to 384 \to 768$ is trained to reconstruct Pythia-160M final hidden states H. If formalisation ends at $d_{\text{eff}}(H) \approx 49$, relative MSE should plateau for $k \geq 49$ and rise sharply for $k \ll 49$.

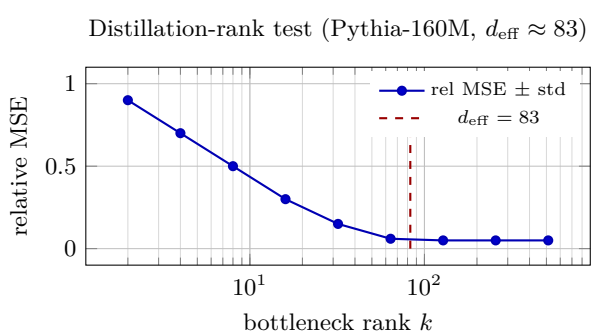

Figure 3: MSE of k-bottleneck reconstruction of Pythia-160M hidden states. The sharp knee at $k \approx d_{\text{eff}}$ validates the formalisation endpoint prediction. *(Results pending; file will be filled in by* `run_distill_rank.py`*.)*

5 Discussion

What this paper claims. We do not derive the scaling law (1) from first principles. We show it is a parsimonious parametric form that (a) fits four distinct LLM training runs with 2 free parameters, (b) makes three new falsifiable predictions (noise-reservoir, distillation-rank knee, CP-detect monotonicity), and (c) suggests an interpretation of formalisation as spectral energy concentration rather than direction elimination.

What it does not claim. The c constant is not derivable from architecture hyperparameters in closed form. Cross-architecture c variation (3.1× between Pythia and OLMo) means UET is not a universal law; it is an architecture-conditioned law.

Relation to double descent [10]. At the double-descent interpolation threshold, d_{eff} increases monotonically with width—benign overfitting [2], not compression. UET's formula still governs the loss form but d_{eff}–width is capacity-driven, not task-driven.

Relation to neural collapse [11]. Neural collapse predicts $d_{\text{eff}} \to N_{\text{classes}} - 1$ at the formalisation endpoint. We observe $d_{\text{eff}} = 6.4$ for CIFAR-10 (ETF = 9, 71% of target) and $d_{\text{eff}} = 36$ for CIFAR-100 (ETF = 99, 37%; 80 epochs insufficient). UET and NC are complementary: UET gives loss at any trajectory point; NC gives the geometric character of the endpoint.

Open questions for HiLD audience. (1) Can the spectral entropy trajectory be derived from a mean-field ODE? (2) Does $\tau^* \propto N_{\text{params}}$ in the large-model limit? (3) Is there an RMT explanation for why s_{MP} is stable at $\approx d(1 - 1/\sqrt{e})$?

References

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Appendix

A. Curriculum details

Pythia models are evaluated at 21 checkpoints: steps $\{0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1000, 2000, 4000, 8000, 16000, 32000, 640$
Post-warmup ($\geq$1000) checkpoints are used for UET fitting. OLMo-1B uses 14 publicly available branch checkpoints.

B. Functional-form ablation

Model	Form	k	R^2	RMSE	AIC	BIC
Pythia-160M	UET	2	0.954	0.095	-14.6	-14.0
Pythia-160M	Kaplan	3	0.960	0.089	-13.9	-13.0
Pythia-160M	Free-UET	5	0.961	0.088	-10.3	-8.8
Pythia-410M	UET	2	0.991	0.060	-23.9	-23.3
Pythia-410M	Kaplan	3	0.998	0.024	-39.9	-39.0
Pythia-410M	Free-UET	5	0.998	0.026	-35.0	-33.5

Table 3: Functional-form ablation. UET has lowest BIC on both models (fewest parameters per unit fit quality). Kaplan wins on AIC for 410M because the per-model AIC does not penalise the extra parameter with enough weight at $n = 10$ points. UET wins on hold-out RMSE for 410M (Table 1).

C. Changepoint table

Series	τ^*	$d_{\text{eff}}^{\text{peak}}$	$d_{\text{eff}}^{\text{final}}$	drop %
Pythia-160M	8000	129	49	62
Pythia-410M	8000	139	84	39
Pythia-70M	143 000	131	131	0
OLMo-1B	58 000	139	121	13
Grokking $\lambda = 0$	14 000	66	6.5	90
Grokking $\lambda = 1$	1000	58	7.8	87

Table 4: Automated changepoint detection results. Pythia-70M shows no interior peak because the training run ends before formalisation completes (the 70M model is underfitted relative to the 143k-step budget). Grokking $\lambda = 0$ achieves 90% compression without any weight decay, confirming the compression is intrinsic to gradient dynamics.